

QAOA for Portfolio Optimisation

FYS5419 Quantum computing
and quantum machine learning

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Abstract

This is an abstract.

1 Introduction

2 Methods

2.1 Mean-Variance Portfolio Optimisation

Financial Data $\rightarrow (\mu, \Sigma) \rightarrow$ QUBO $\rightarrow$ Ising Hamiltonian $\rightarrow$ Pauli operators $\rightarrow$ QAOA

$$\min_x (-\mu^\top x + \lambda x^\top \Sigma x) \quad (2)$$

$-\mu^\top x$ maximises expected return and $\lambda x^\top \Sigma x$ penalises portfolio variance (risk).

$$\sum_{i=1}^n x_i = K \quad (3)$$

$$C(x) = -\mu^\top x + \lambda x^\top \Sigma x + A \left(\sum_{i=1}^n x_i - K \right)^2 \quad (4)$$

where $A > 0$ is a penalty coefficient chosen large enough that the minimum of $C(x)$ always satisfies the budget constraint.

$$r_i(t) = \frac{P_i(t+1) - P_i(t)}{P_i(t)} \approx \ln \frac{P_i(t+1)}{P_i(t)} \quad (5)$$

$$\mu_i = \langle r_i \rangle_t \quad \Sigma_{ij} = \langle r_i r_j \rangle - \langle r_i \rangle \langle r_j \rangle \quad (6)$$

2.2 Quadratic Unconstrained Binary Optimisation (QUBO)

QUBO formulation

$$\min_{x \in \{0,1\}^n} C(x) = \sum_i Q_{ii} x_i + \sum_{i < j} Q_{ij} x_i x_j \quad (7)$$

where for binary variables $x_i^2 = x_i$

$$C(x) = \sum_i [-\mu_i + \lambda \Sigma_{ii} + A(1 - 2K)] x_i + \sum_{i < j} [2\lambda \Sigma_{ij} + 2A] x_i x_j + AK^2 \quad (8)$$

$$C(x) = \sum_i \underbrace{[-\mu_i + \lambda \Sigma_{ii} + A(1 - 2K)]}_{Q_{ii}} x_i + \sum_{i < j} \underbrace{[2\lambda \Sigma_{ij} + 2A]}_{Q_{ij}} x_i x_j + AK^2 \quad (9)$$

$$\boxed{Q_{ii} = -\mu_i + \lambda \Sigma_{ii} + A(1 - 2K) \quad Q_{ij} = 2\lambda \Sigma_{ij} + 2A \quad (i > j)} \quad (10)$$

2.3 Ising Hamiltonian Mapping

Mapping $x_i \in \{0, 1\}$ to spin variables $z_i \in \{-1, +1\}$ via

$$x_i = \frac{1 - z_i}{2} \quad (11)$$

Substituting into $C(x)$ and collecting terms by degree giving and Ising Hamiltonian.

$$H(z) = \sum_i h_i z_i - \frac{1}{4} \sum_{j \neq i} Q_{ij} \quad J_{ij} = \frac{Q_{ij}}{4} \quad (i < j) \quad (12)$$

and the total additive constant is

$$c = \text{offset} + \frac{1}{2} \sum_i Q_{ii} + \frac{1}{4} \sum_{i < j} Q_{ij} \quad (13)$$

Such that $H(z) + c = C(x) \quad \forall x$

2.4 Quantum Pauli Operators and Cost Hamiltonian

$z_i \in \{-1, +1\}$

$$z_i \longrightarrow \hat{Z}_i = I^{\otimes(i)} \otimes Z \otimes I^{\otimes(n-i-1)} \quad (14)$$

$$Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad (15)$$

has eigenvalues ± 1

The costing Hamiltonian is then the $2^n \times 2^n$ matrix

$$\boxed{H_c = \sum_i h_i \hat{Z}_i + \sum_{i < j} J_{ij} \hat{Z}_i \hat{Z}_j} \quad (16)$$

Every computational basis state $|z\rangle$ is an eigenstate of H_c with eigenvalue $H(z) = \sum_i h_i z_i + \sum_{i<j} J_{ij} z_i z_j$. The mixing Hamiltonian drives quantum tunneling between bistring states

$$H_m = \sum_{i=j}^n \hat{X}_i \quad X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad (17)$$

2.5 QAOA Algorithm

Single-layer $p = 1$ ansatz

$$|\psi(\gamma, \beta)\rangle = e^{-i\beta H_m} e^{-i\gamma H_c} |+\rangle^{\otimes n} \quad (18)$$

Where $e^{-i\gamma H_c}$ is the phase separation unitary, it phases each basis state by $e^{-i\gamma H(z)}$ encoding cost information into amplitudes.

$e^{-i\beta H_m}$ is the mixing unitary, it spreads amplitudes between states, allowing transitions between candidate solutions.

$$\underbrace{H^{\otimes n}}_{\text{superposition}} \longrightarrow \underbrace{e^{i\gamma H_c}}_{\text{phase separation}} \longrightarrow \underbrace{e^{-i\beta H_m}}_{\text{mixing}} \longrightarrow \underbrace{\text{measure}}_{z\text{-basis}} \quad (19)$$

Multi-layer $p > 1$ ansatz, for depth p , alternate p cost, and p mixer layers

$$|\psi(\gamma, \beta)\rangle = \prod_{k=1}^p (e^{-i\beta_k H_m} e^{-i\gamma_k H_c}) |+\rangle^{\otimes n} \quad (20)$$

With $2p$ variational parameters $(\gamma_1, \dots, \gamma_p, \beta_1, \dots, \beta_p)$

The classical optimiser minimises the energy expectation value

$$E(\gamma, \beta) = \langle \psi(\gamma, \beta) | H_C | \psi(\gamma, \beta) \rangle \quad (21)$$

Since H_C is diagonal with eigenvalues $H(z)$ this equals

$$E = \sum_{z \in \{-1, +1\}^n} H(z) |\langle z | \psi \rangle|^2 \quad (22)$$

2.6 Code

- Set parameters: λ, A, K
- Build the QUBO matrix: Q
- Convert to Ising coefficients: (h, J, c)
- Build the quantum operators H_C and H_m
- Run brute-force classical search (except for validation)
- Run QAOA for $p = 1, 2, 3$ and compare

2.7 Statistical Physics Connection

The Ising Hamiltonian

$$H(z) = \sum_i h_i z_i + \sum_{i<j} J_{ij} z_i z_j \quad (23)$$

defines a classical partition function at inverse temperature β_{th}

$$Z = \sum_{\{z\}} e^{-\beta_{th} H(z)} \quad (24)$$

and the associated free energy

$$F = -\frac{1}{\beta_{th}} \ln Z \quad (25)$$

QAOA as variational free-energy minimisation, can be understood as a variational approximation of the ground state of H_c

$$E(\gamma, \beta) = \langle \psi | H_C | \psi \rangle \geq E_0 \quad (26)$$

2.8 Quantum Neural Network for Portfolio Optimisation

For asset i with feature vector $\phi_i \in \mathbb{R}^F$ and trainable weight matrices $W^Y, W^Z \in \mathbb{R}^{L \times F}$ the circuit is

$$|\psi_i\rangle = \prod_{t=1}^L R_Z((W_t^Z)^\top \phi_i) R_Y((W_t^Y)^\top \phi_i) |0\rangle \quad (27)$$

where each layer encodes the data afresh via the rotations

$$R_Y(\theta) = \begin{pmatrix} \cos \frac{\theta}{2} & -\sin \frac{\theta}{2} \\ \sin \frac{\theta}{2} & \cos \frac{\theta}{2} \end{pmatrix} \quad R_Z(\theta) = \begin{pmatrix} e^{-i\frac{\theta}{2}} & 0 \\ 0 & e^{i\frac{\theta}{2}} \end{pmatrix} \quad (28)$$

The feature vector is augmented with a bias term $\phi_i = (\mu_i^{sc}, \sigma_i^{sc}, 1)$ so the weight matrix also plays the role of trainable offsets. The circuit output is the Z -expectation:

$$s_i = \langle \psi_i | Z | \psi_i \rangle = |\langle 0 | \psi_i \rangle|^2 - |\langle 1 | \psi_i \rangle|^2 \in [-1, +1] \quad (29)$$

Differentiable portfolio selection, to connect the scores to a differentiable portfolio cost, we use a sigmoid-based soft ranking. Find a threshold t such that

$$p_i = \sigma(\alpha(s_i - t)) \quad \sum_i p_i = K \quad (30)$$

where $\sigma(x) = (1 + e^{-x})^{-1}$ and α is a sharpness parameter. The threshold is found by bisection. The continuous selection probabilities $p_i \in (0, 1)$ then enter the mean-variance cost:

$$\mathcal{L}(W^Y, W^Z) = -\mu^\top \mathbf{p} + \lambda \mathbf{p}^\top \Sigma \mathbf{p} + \left(\sum_i p_i - K \right)^2 \quad (31)$$

Parameter shift-rule

$$\frac{\partial s_i}{\partial W_{lf}^Y} = \frac{s_i(W_{lf}^Y + \frac{\pi}{2}) - s_i(W_{lf}^Y - \frac{\pi}{2})}{2} \quad (32)$$

The ADAM optimiser

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) \nabla \mathcal{L} \quad (33)$$

$$v_t = \beta_2 v_{t-1} + (1 - \beta_2) (\nabla \mathcal{L})^2 \quad (34)$$

$$W \leftarrow W - \eta \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \varepsilon} \quad (35)$$

With bias-corrected moments

$$\hat{m}_t = \frac{m_t}{(1 - \beta_1^t)} \quad (36)$$

$$\hat{v}_t = \frac{v_t}{(1 - \beta_2^t)} \quad (37)$$

3 Results

4 Conclusion

5 Appendix

Table 1. Influential Figures and Their Seminal Contributions to Quantum Computing

Figure	Field	Most Famous Work / Contribution
Max Planck	Quantum Mechanics	Discovery of energy quanta ($E = h\nu$)
Niels Bohr	Quantum Mechanics	Bohr model of the atom and complementarity
Werner Heisenberg	Quantum Mechanics	Uncertainty principle and matrix mechanics
Erwin Schrödinger	Quantum Mechanics	Schrödinger equation and wave mechanics
Richard Feynman	QM / Computing	Path-integral formulation and quantum simulators
John Bell	Quantum Foundations	Bell’s theorem (violation of local realism)
Peter Shor	Quantum Algorithms	Shor’s factoring algorithm (exponential speedup)
Lov Grover	Quantum Algorithms	Grover search (quadratic speedup for unstructured search)
Maria Schuld	Quantum ML	Quantum kernels and variational quantum circuits
Edward Farhi	QAOA	Quantum Approximate Optimization Algorithm

Table 2. QAOA Circuit Components

Component	Notation	Description
Initial state	$ +\rangle^{\otimes n}$	Uniform superposition prepared by Hadamard gates on all n qubits
Cost unitary	$U_C(\gamma)$	Phase rotations proportional to the QUBO objective; encodes problem structure
Mixer unitary	$U_B(\beta)$	$R_X(2\beta)$ rotations on each qubit; drives exploration of solution space
Ansatz state	$ \psi(\gamma, \beta)\rangle$	Alternating product of p cost and mixer layers
Cost function	$F_p(\gamma, \beta)$	Expectation value of the cost operator C ; minimized classically
Approximation ratio	$r = F_p/F_{\text{opt}}$	Quality metric relative to exact brute-force optimum